

Multi-domain Fake News Detection for History News Environment Perception

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Abstract—As social networks continue to grow, fake news in different areas on social media is spreading widely. Numerous parts of the actual world, such as politics and daily living, are threatened by fake news. Most traditional fake news detection focuses on a single domain, so identifying multi-domain fake news is receiving increasing attention. To improve the detection of multi-domain fake news, a novel multi-domain fake news detection model is proposed. We created the History News Environment Perception framework by analyzing the relationship between domain news and its news environment sample (HNEP). The approach is that for each post, we build the history and mainstream news environment from all previous posts. We design a perception module for mainstream and history news environments to improve multi-domain fake news prediction by fusing perceptual representations via domain gates. The experimental results demonstrate that HNEP can enhance the performance of multi-domain fake news detection.

Index Terms—fake news, multi-domain, HNEP

I. INTRODUCTION

Since 2006, microblogs have emerged as one of the main sources of news and information for users due to the rapid development of the Internet. Social media has brought convenience to people, but it has also created opportunities for fake news to spread. Weibo has 582 million monthly active users as of March 2022, and 56% of Internet users get information from platforms like Weibo. Fake news spreads fast and widely, and according to Weibo's 2021 annual report on disinformation, 66,251 fake news items were successfully addressed. The spread of fake news can be harmful and influence society negatively, and identifying fake news also requires a lot of human and financial resources. Therefore, automatic detection of fake news [1] is an important problem that must be solved immediately.

On a daily basis, social media platforms generate a large amount of news from different domains. However, most existing research [2,3,4,5] focuses only on posts featuring a single domain, e.g., COVID-19, Politifact. Liang et al [1] used user behavior features to model. Qazvinian et al [5] use a feature-specific enhancement model based on content features and Weibo platform. Sun et al [7] proposed fusing topic features and sentiment features fed into a fake news

detector. Fake news propagation features tend to work well when applied to single-domain detection, but are less effective in detecting other domains. In reality, news that is shared on social media typically comes from multiple domains, making multi-domain false news incredibly difficult due to the lack of labeled data for some domains.

However, the data distribution of multi-domain news is not uniform and there is a problem with domain shift [8]. Different fields use different words, such as "epidemic", "symptom" in medical and health news, "casualty" and "catastrophe" in disaster news. To address the domain shift, Silva et al [9] reduced the domain differences by multimodal signals. Nan et al [10] proposed using nine expert networks to extract representations of multiple domains and then enhance the fake news detection performance by domain gate fusion. But these approaches ignore the external news environment in which fake news is created and disseminated. Sheng et al [11] constructed a News Environment Signal Perception Module (NEP) by publishing recent mainstream news with fake news. Augmenting fake news detection with NEP is proven to be effective, but has one drawback. Fake news is often reprocessed on widely-followed topics, and its news environment is related to recent mainstream news. However, when fake news is altered from historical hot news, the current mainstream news signal perception will be invalidated.

This research proposes a framework for fusing perceptions of history and external mainstream news environments (HNEP). For each post, we identify meaningful signals from both the history news environment and the external mainstream news environment to compensate for the weak correlation with the news environment, resulting in a more accurate categorization and enhanced performance of multi-domain fake news.

The following are the important innovations and contributions of this paper:

- To capture valuable signals through news, our framework for fusing the perception of the external news environment and history news environment improves the

association between news and the news environment.

- We provide a novel multi-domain fake news framework (HNEP-MDFEND) that enhances fake news detection in the majority of domains.
- We have conducted extensive experiments on publicly available multi-domain datasets, and the experimental results show that the proposed framework can effectively improve the performance of multi-domain fake news, outperforming other models in all metrics.

II. RELATED WORK

Early fake news detection reduces its harm. Researchers have two main tasks to defend against fake news in different fields: 1) Single-Domain Fake News Detection (SFND): In conventional research, complex manual features are constructed and fed to machine learning classifiers such as SVM [12], K-means [13], etc. Recent research has combined the propagation characteristics of fake news with the characteristics of release time to develop models. Ma et al [14] used dynamic temporal structure (DSTS) to obtain representations. Yu et al [15] separated news by time period to obtain text features, which were then fed into CNN to learn local representations of temporal sequences. Other works [16,17] combine text and images for multimodal feature modeling. Wang et al [18] eliminate the tight dependence on domain-specific events by employing event adversarial training. However, they did not use information from outside the field, such as the history news environment or the mainstream news environment. 2) Multi-domain Fake News Detection: The primary objective of multi-domain fake news detection is to determine the veracity of news in several domains, such as catastrophe and accident reporting and social life. However, the number of fake news in the fields of education, examination, finance, and business is significantly insufficient, rendering the detection models ineffective. Analyzing prior studies [19,20], we discovered that multi-domain false news detection is effective in different domains with a significant number of samples but has larger limits when the sample size is limited. Consequently, this paper proposes a framework for fusing the history news environment and the mainstream news environment (HNEP), which can compensate for the small sample size of single domain news and serve as a new domain event feature to capture both the external news environment and the history news environment, and thus identify fake news more effectively.

III. OUR PROPOSED FRAMEWORK

In order to perceive the history news environment and the mainstream news environment more effectively, we designed the framework HNEP. Given a post p , we construct its history macro and micro environments (HISTORY MACROENV and HISTORY MICROENV) using all news before the posting event. Meanwhile, we construct the mainstream news environment, including macro and micro environments (MAINSTREAM MACROENV and MAINSTREAM MICROENV) for posting recent news. Then we model it individually to obtain the perception vectors of the history environment

$\mathbf{V}^{p,history\ mac}$, $\mathbf{V}^{p,history\ mic}$ and the perception vectors of the mainstream environment $\mathbf{V}^{p,main\ mac}$ and $\mathbf{V}^{p,main\ mic}$.

Finally, the history environment perception vector and the mainstream news environment perception vector are merged with the generated post representation o of the fake news detector to predict the veracity of p . The overall structure of the network model is shown in Fig. 1.

A. Input Representation

Each post and the news environment text are encoded using the pre-trained language model BERT [21], and the news environment posts are post-trained using SimCSE [22]. The initial representation is the BERT output for the token.

B. News Environment Construction

History News is a collection of posts that are most similar to the current post and all previously published news. Typically, official media channels with millions of followers disseminate mainstream news items that are intended for a general audience.

Let ε be the set of all news that predates the p release event. For the history news environment (HISTORY MACROENV) and the mainstream news environment (MAINSTREAM MACROENV), we create a macro-environment and a micro-environment, which are defined as follows:

- HISTORY MACROENV is a collection of all news items published before the release of p ; MAIN STREAM MACROENV includes a collection of all news from the past T days.

$$\mathcal{E}^{history\ mac} = \{e : e \in \mathcal{E}, 0 \leq t_e \leq t_p\} \quad (1)$$

$$\mathcal{E}^{main\ mac} = \{e : e \in \mathcal{E}, 0 < t_p - t_e \leq T\} \quad (2)$$

Where t_p and t_e respectively denote the release date of p and the news e .

- HISTORY MICROENV and MAINSTREAM MICROENV are the news sets that are most relevant to p in $\mathcal{E}^{history\ mac}$ and $\mathcal{E}^{main\ mac}$.

$$\mathcal{E}^{mic} = \{e : e \in \text{Topk}(p, \mathcal{E}^{mac})\} \quad (3)$$

Where the proportion is determined by the value of $k = \lceil r |\mathcal{E}^{mac}| \rceil$ and $r \in (0, 1)$.

In general, the history and mainstream macro environment provide a macro perspective on what the general public has historically focused on and what has lately garnered attention, whereas the history and mainstream micro environment depict the distribution of items for similar events.

C. History news and mainstream news environmental perception

The purpose of news environment perception is to get useful signals from history news and mainstream news. From the motivation of fake news creators to publish, we let the model perceive the MACROENV and MICROENV of

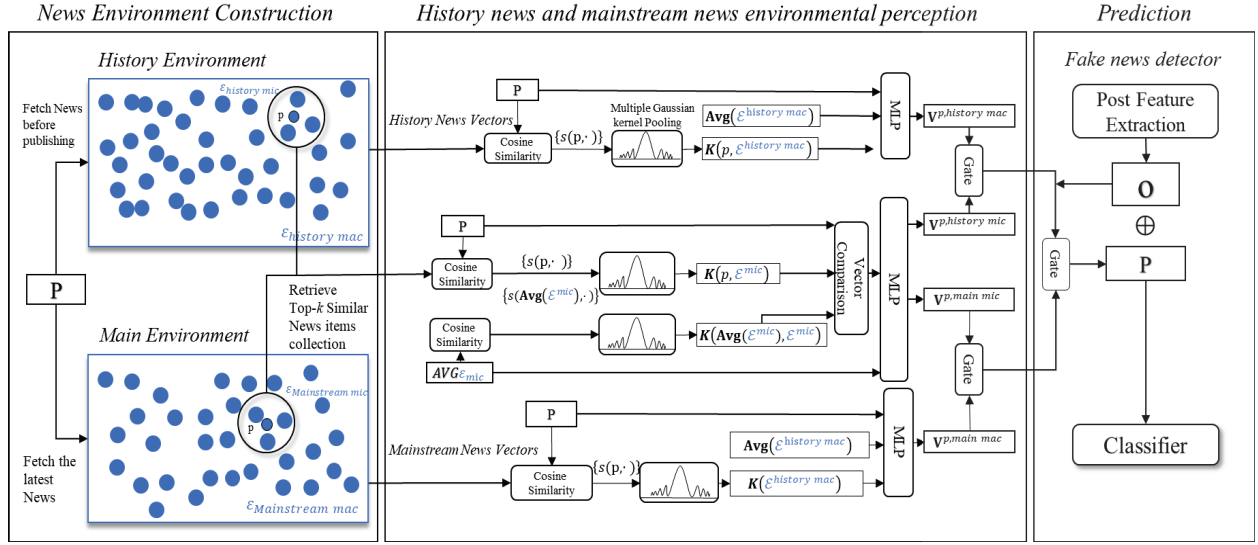


Fig. 1. The architecture of History News Environment Perception (HNEP) for Multi-domain fake news detection

mainstream news. We can deduce from the MACROENV perception module that fabricated news is more likely to propagate and have more influence. Following [11], we first present the perception of the history news environment, in which we translate the degree of p 's popularity in history news and mainstream news into similarity estimates for p and specific news items. In short, if p is comparable to numerous stories in the history news macroenvironment, then it is popular in that environment and more likely to garner public interest.

$$s(p, e_i) = \frac{p \cdot e_i}{\|p\| \|e_i\|} \quad (4)$$

Since the lengths of $\{\cos(p, e_i)\}_{i=1}^{|\mathcal{E}^{history\ mac}|}$ and $\{\cos(p, e_i)\}_{i=1}^{|\mathcal{E}^{main\ mac}|}$ are uncertain. As a result, we use Gaussian kernel pooling to get soft counts and c kernels to calculate similarity on a given input dimension. As a starting point, we choose 21 kernels with σ_k^2 of 0.05 and a stride scatter in $[-1, 1]$ of 0.1. After the K th, the kernel's output is:

$$K_k^i = \exp\left(-\frac{(s(p, e_i) - \mu_k)^2}{2\sigma_k^2}\right) \quad (5)$$

$$K_k(p, \mathcal{E}^{history\ mac} | \mathcal{E}^{main\ mac}) = \sum_{i=1}^{|\mathcal{E}^{mac}|} K_k^i \quad (6)$$

Then we get the sum by normalizing all kernel's output. where $Norm(\cdot)$ denotes normalization.

$$K(p, \mathcal{E}^{history} | \mathcal{E}^{main\ mac}) = Norm\left(\sum_{k=1}^C K_k(p, \mathcal{E}^{mac})\right) \quad (7)$$

We obtained the popularity scores for p and the history and mainstream news contexts by calculating $K(p, \mathcal{E}^{history} | \mathcal{E}^{main\ mac})$. To augment the perceptual information, we aggregate the mean vectors of macro environments using MLP to produce perceptual vectors ($v^{p, history\ mac}, v^{p, main\ mac}$) of history and mainstream news environments for p .

$$v^{p, mac} = MLP(p \oplus AVG(\mathcal{E}^{mac}) \oplus K(p, \mathcal{E}^{mac})) \quad (8)$$

For MICROENV, it contains the most pertinent news in the macro environment with p . Given that fake news is more novel than actual news, the likelihood that a post is false is significantly raised if it is novel. A novel perspective that can provide news contained in the micro environment. We again use Eqs. (5)-(10) to obtain their microenvironmental perception vectors $v^{p, history\ mic}, v^{p, main\ mic}$. Simultaneously, we use $K(p, \mathcal{E}^{mic}), K(AVG(\mathcal{E}^{mic}), \mathcal{E}^{mic})$, which are aggregated using MLP, to improve the micro perception ability. The calculation for p 's HISTORY MICROENV and MAIN MICROENV is as follows:

$$w^{semantic} = MLP(p \oplus AVG(\mathcal{E}^{mic})) \quad (9)$$

$$w^{sim} = MLP(f(K(p, \mathcal{E}^{mic}), K(AVG(\mathcal{E}^{mic}), \mathcal{E}^{mic}))) \quad (10)$$

The comparison function f indicates that the vector is concatenated by the difference between the Hadamard Product and the vector. w^{semantic} and w^{sim} , respectively, include semantic and similarity information. Finally they are fed into MLP aggregation to produce $v^{p, \text{historymic}}$ and $v^{p, \text{mainmic}}$.

D. Prediction

We use the domain gate mechanism to adaptively fuse the macro-perception vector and the micro-perception vector of history and mainstream news, and then we use the domain gate mechanism to adaptively fuse the history and mainstream perceptions to produce the final environmental perception vector.

$$v^h = g \odot v^{p, \text{history mac}} + (1 - g) \odot v^{p, \text{history mic}} \quad (11)$$

$$v^m = g \odot v^{p, \text{main mac}} + (1 - g) \odot v^{p, \text{main mic}} \quad (12)$$

$$v^p = g \odot v^h + (1 - g) \odot v^m \quad (13)$$

where the gating vector $g = \text{sigmoid}(\text{MLP}(\mathbf{o} \oplus v^p))$, and $\mathbf{o}$ represents the last-layer characteristic of a post-only detector. MLP aggregates $\mathbf{o}$ and v^p and feeds into Softmax for prediction.

$$\hat{y} = \text{softmax}(\text{MLP}(\mathbf{o} \oplus v^p)) \quad (14)$$

We train our model to minimize Binary Cross-Entropy Loss(BCELoss). $\hat{y}$ be the distribution of fake and real news prediction labels.

$$L = - \sum_{i=1}^N (y^i \log \hat{y}^i + (1 - y^i) \log (1 - \hat{y}^i)) \quad (15)$$

IV. EXPERIMENTAL RESULTS

A. Hyperparameters Setting

Our applies the proposed framework to multi-domain fake news detection. In our experiments, BERT is used to initialize all word vector weights. T set to 3 days. Particularly, we shall truncate the first 500 samples as the macroenvironment, take the Top 20% as the microenvironment. Word vector dimension is 768, MLP dimension is 128, learning rate is 5e-5, batchsize set to 64, epoch 50, and early-stop set to 10.

B. DataSets

The multi-domain Weibo news dataset WEIBO 21[10], which covers true and fake news on Sina Weibo from December 2014 to March 2021, is the subject of this study's experiments. The release time, real and fake news, field labels, and maps are included with each news item. The distribution of datasets is shown in Table I.

The news environmental items that we utilize are compiled by [11]. It contains a total of 583,208 news stories produced between 2010 and 2021 by media outlets with over 10 million readers, such as People's Daily, Xinhua Agency, and CCTV News.

TABLE I
DATA DISTRIBUTION OF WEIBO21

Domain	Science	Military	Education	Disasters	Politics
real	143	121	243	185	306
fake	93	222	248	591	546
all	236	343	491	776	852
Domain	Health	Finance	Entertainment	Society	ALL
real	485	959	1000	1198	4640
fake	515	362	440	1371	4488
all	1000	1321	1440	2669	9128

C. Model comparisons

Finally, we use the F1-Macro score metric to measure the performance of each domain for fake news. Also, to validate the performance of our model, we compared it with some baseline models. We list them as follows:

- TextCNN[24]: We directly model by the usage of TextCNN. Word2vec is responsible for embedding all of TextCNN's input characteristics.
- BERT[21]: First, the pre-training settings of Bert are loaded into the model. Predicts the real or fake of multi-domain fake news directly after BERT fine-tuning.
- EANN[23]: It eliminates event-specific properties using a strategy known as event adversarial, and its model architecture is identical to that of TextCNN.
- EDDFN[25]: It is a model for cross-domain fake news detection that models different domains while retaining domain-specific knowledge, here we use manually labeled ones for fake news detection.
- NEP[11]: It builds a news environment perception framework to enhance fake news detection.
- MDFEND[10]: It uses multiple expert network distributions to extract domain representations.

The experimental results of the above baseline models are shown in Table II. Overall, the results of our proposed framework in most domains outperform these baseline models.

D. Case Study

Table II shows the comparative results of HNEP against several baselines. In the majority of domains, HNEP- MD-FEND surpasses all baseline models. We discovered that the TextCNN and BERT classification models are less successful since they often focus on a single domain and perform poorly when applied to multi-domain information. In contrast, we have enhanced the performance of the Science and Technology, Military, Disasters, Politics, Finance, and Social domains in the multi-domain model. HNEP enhances its multiple domain features by generating the history news environment and the mainstream news environment concurrently, and solves the problem of NEP framework's over-reliance on mainstream news, which is superior to the soft combination of multiple domain expert extraction domain representation (MDFEND). Therefore, the HNEP-MDFEND model performs better than other multi-domain models.

TABLE II
COMPARISON RESULTS

Domain	Science	Military	Education	Disasters	Politics	Health	Finance	Entertainment	Society	ALL
TextCNN	0.7524	0.8839	0.8362	0.8222	0.8561	0.8768	0.8638	0.8456	0.8540	0.8686
BERT	0.7777	0.9072	0.8331	0.8512	0.8366	0.9090	0.8735	0.8769	0.8577	0.8795
EANN	0.8225	0.9274	0.8624	0.8666	0.8705	0.9150	0.871	0.8957	0.8877	0.8975
EDDFN	0.8286	0.9137	0.8676	0.8789	0.8478	0.9379	0.8636	0.8832	0.8689	0.8919
NEP	0.8333	0.8821	0.8384	0.8785	0.9111	0.8653	0.8985	0.9096	0.8927	0.8990
MDFEND	0.8301	0.9389	0.8917	0.9003	0.8865	0.9400	0.8951	0.9066	0.8980	0.9137
HNEP-MDFEND	0.9366	0.9598	0.8686	0.9159	0.9351	0.9225	0.8967	0.9158	0.9208	0.9255

V. CONCLUSIONS

In this paper, we propose the HNEP framework for multi-domain fake news detection. Our method operates by fusing the perception of the historical news environment and the perception of the mainstream news environment as the environment perception vector of posts, which is then adaptively fused by domain gates and sent to the fake news detector.

To evaluate the performance of the proposed algorithm, compare the F1-Score of five algorithms on a Multi-domain fake news dataset. Experimental results manifest that the HNEP framework can effectively enhance the performance of multi-domain fake news detection.

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